

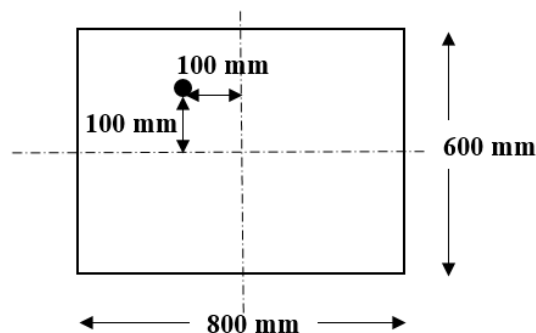
CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Fourth Semester of B.Tech.Theory Examination (CIVIL)****APRIL – MAY 2018****CL 209 STRUCTURAL ANALYSIS- I****Date: 04.05.2018 (Friday)****Time: 10:00 a.m. to 01:00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat sketch wherever required.
4. Use of scientific calculator is allowed.

SECTION - I**Q - 1 Do as directed.**

- i) Sketch the resultant stress distribution at the base section for condition that direct stress is equal to bending stress, greater than bending stress & less than bending stress. **[03]**
- ii) If the crushing stress in the material of a mild steel column is 330 MPa and modulus of elasticity $E = 2.1 \times 10^5 \text{ N/mm}^2$ then Euler's formula for critical load is applicable for a slenderness ratio equal to or greater than
(a) 40 (b) 50 (c) 60 (d) 80 **[01]**
- iii) For a circular column with its ends hinged, the slenderness ratio is 160. The L/d ratio of the column is
(a) 80 (b) 57 (c) 40 (d) 20 **[01]**
- iv) To determine critical load, Rankine's formula is applicable when **[01]**
(a) the column is short (c) the column is long
(b) the column is medium (d) all of the above.
- v) Define crippling load. **[01]**

- Q - 2 (a)**
- i) Find out the limit of eccentricity for solid circular section of diameter 'D' and draw the core of the section. **[02]**
 - ii) A column 800 mm x 600 mm is subjected to an eccentric load of 60 kN as shown in **figure 1**. Calculate maximum and minimum intensity of stresses in the column. **[05]**

**Figure 1 [Q-2 (b)]**

- Q - 2 (b)** i) Write down Rankine's formula for critical load of a column and state the meaning of all notations used in the formula. [02]
- ii) A solid rod 250 cm long and 6 cm in diameter is used as a column whose one end is fixed and other end is hinged. If $E = 2 \times 10^4 \text{ kN/cm}^2$ and factor of safety = 4, determine the safe compressive load for this column. [05]

OR

- Q - 2 (b)** i) Explain the failure of different types of column. [03]
- ii) Derive the formula for the Euler's crippling load of a column having both ends hinged. [04]

Q - 3 Answer any two questions. [14]

- (a) Draw qualitative influence line diagram for reaction at A, reaction at B, shear at section 3 m from left support and bending at section 3 m from left support for the simply supported beam of span 10 m. Also show the ordinates at important location.
- (b) Explain different types of rolling load with suitable example. Also explain the difference between influence line diagram and bending moment diagram.
- (c) An u.d.l. of 12 kN/m and 3 m length crosses a simply supported girder of span 10 m from left to right. Draw influence line for S.F. and B.M. at section 4 m from left support and find maximum S.F. and B.M. at this section.

SECTION – II

- Q - 4 (a)** Explain determinate and indeterminate structure with an example. [03]
- Q - 4 (b)** i) Calculate Static Indeterminacy and Kinematic Indeterminacy for the following plane truss. **Refer figure 2.** [02]

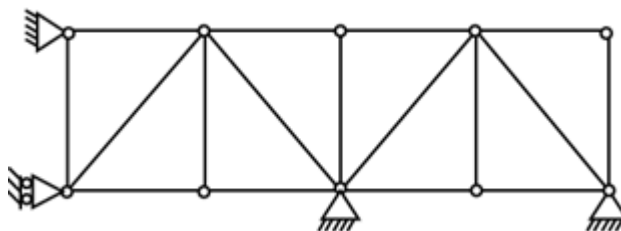


Figure 2 [Q-4 (b) (i)]

- ii) Calculate Static Indeterminacy and Kinematic Indeterminacy for the following plane frame. **Refer figure 3.** [02]

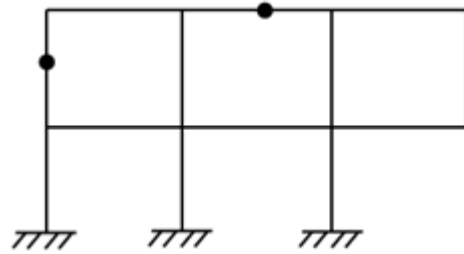


Figure 3 [Q-4 (b) (ii)]

Q - 5 Attempt any two.

[14]

- a) Calculate slope at 'B' θ_b and deflection at 'B' δ_b for the cantilever beam shown in **figure 4** using Moment area method OR Conjugate beam method. Consider $E = 2 \times 10^5 \text{ N/mm}^2$ and $I = 5 \times 10^8 \text{ mm}^4$.

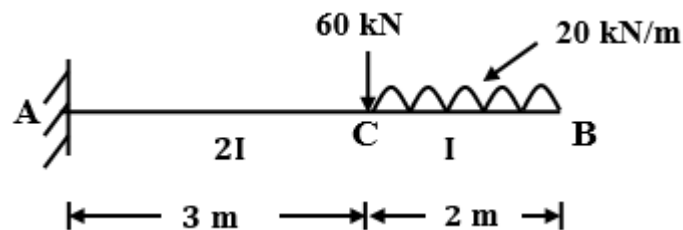


Figure 4 [Q-5 (a)]

- b) Calculate deflection at 'C' δ_c and deflection at 'D' δ_d for the simply supported beam shown in **figure 5** using Double integration method. Consider $E = 2 \times 10^5 \text{ N/mm}^2$ and $I = 3 \times 10^8 \text{ mm}^4$.

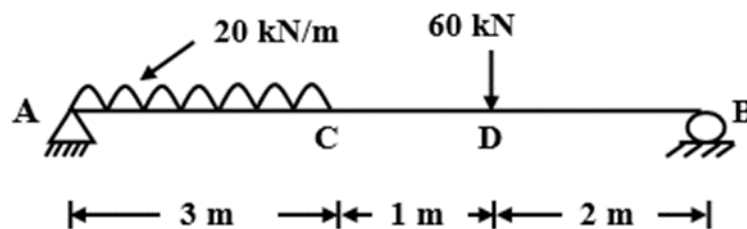


Figure 5 [Q-5 (b)]

- c) Calculate the slope and deflection at free end of cantilever beam shown in **figure 6** by conjugate beam method. Take $E = 2 \times 10^5 \text{ N/mm}^2$ and $I = 4 \times 10^5 \text{ m}^4$.

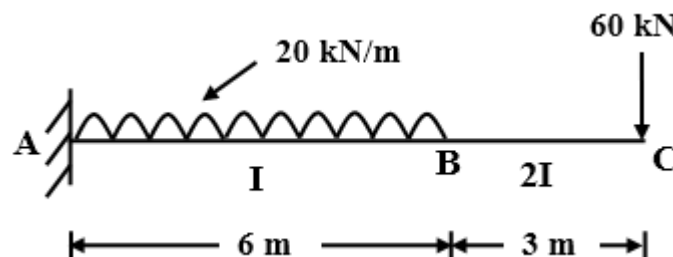


Figure 6 [Q-5 (c)]

- Q - 6 (a)** A three hinged parabolic arch has span 36 m and central rise of 8 m. It carries uniformly distributed load of 4 kN/m over left half of the span. Calculate the reactions at both ends. Also calculate the bending moment, normal thrust and radial shear at 9 m from left hinge. **[09]**

OR

- Q - 6 (a)** A 3 hinged circular arch of 25 m span and 5 m of central rise is loaded with a concentrated force of 80 kN at 7.5 m from the left hinge. Find (i) Support reactions (ii) Maximum positive bending moment in left span and maximum negative bending moment in right span. **[09]**

- Q - 6 (b)** A thin spherical shell of 800 mm diameter is 6 mm thick. It is filled with a liquid, so that the internal pressure is 1 N/mm². Determine the increase in diameter and capacity of the shell. Take $E = 200 \text{ GPa}$ and $\mu = 0.3$. **[05]**

OR

- Q - 6 (b)** A cylindrical vessel 1.5 m in diameter and 4 m long is closed at ends by rigid plates. It is subjected to an internal pressure of 3 N/mm². If the maximum principal stress is not to exceed 150 N/mm², find the thickness of the shell. Assume E as $2 \times 10^5 \text{ N/mm}^2$ and Poisson's ratio as 0.25. Find the changes in diameter, length and volume of the shell. **[05]**
